

Dense Fréchet points for densely continuous quasiconvex functions

A full affirmative answer to the open question in arXiv:1301.2852

11 August 2026

Status. `candidate_full_solution_likely_valid_needs_human_review.`

Abstract

Rabier asks whether every real-valued, densely continuous, quasiconvex function on a separable Banach space is continuous and classically Gâteaux differentiable at the points of a dense subset. We prove that the answer is yes. In the only unresolved region, the interior of the essential infimum sublevel, we obtain the stronger conclusion that Fréchet differentiability points with derivative zero are dense. The new ingredient is a uniform relative-hole lemma for closed convex sets with empty interior.

1 Source question and notation

Let X be a separable Banach space and let $f : X \rightarrow \mathbb{R}$ be real-valued, quasiconvex, and densely continuous. Following [1], put

$$F_\alpha = \{x : f(x) < \alpha\}, \quad F'_\alpha = \{x : f(x) \leq \alpha\},$$

and let

$$m = \mathcal{T}\text{-ess inf}_X f = \sup\{\alpha : F_\alpha \text{ is of first category}\}.$$

The Introduction of [1] asks whether every such f is continuous and classically Gâteaux differentiable at the points of a dense subset; see Figure 1. Its Theorem 9.2 answers only after weakening Gâteaux differentiability to testing a residual set of directions.

The source settles all but the following case. Assume $m > -\infty$ and F'_m is of second category. Source Theorem 3.2(ii) says that

$$V := \text{int } F'_m \neq \emptyset.$$

Since F'_m is convex, V is dense in F'_m . Source Theorem 3.1 says that F_α is nowhere dense whenever $\alpha < m$.

2 Proof intuition

Inside an arbitrary ball in V , avoid the closure of one lower sublevel, then a higher lower sublevel, and so on. Closed convex sets with empty interior leave holes of a fixed relative size: an R -ball always contains an $R/8$ -ball at distance more than $R/4$ from the set. We may therefore nest holes with geometrically shrinking radii. If the selected levels approach m quadratically faster than those radii shrink, their unique common point x satisfies $f(x) = m$, while every nearby y is trapped between $f(x)$ and the appropriate lower level. Dividing this vertical gap by $\|y - x\|$ gives a quantity tending to zero. Thus the construction yields a Fréchet point, not merely directional differentiability.

3 Uniform convex holes

Lemma 1 (Uniform convex-hole lemma). *Let $C \subset X$ be closed and convex with empty interior. Every closed ball $\overline{B}(a, R)$, $R > 0$, contains a closed ball $\overline{B}(b, R/8)$ such that*

$$\text{dist}(w, C) > R/4 \quad \text{for every } w \in \overline{B}(b, R/8).$$

Proof. Because C has empty interior, choose $z \in B(a, R/4) \setminus C$. Strong separation of the point z from the closed convex set C gives $\ell \in X^*$ with $\|\ell\| = 1$ and

$$\ell(z) > \sup_{c \in C} \ell(c).$$

Choose $v \in X$, $\|v\| = 1$, with $\ell(v) > 3/4$, and put $b = z + (R/2)v$. If $\|w - b\| \leq R/8$, then

$$\ell(w) > \ell(z) + \frac{3R}{8} - \frac{R}{8} > \sup_{c \in C} \ell(c) + \frac{R}{4}.$$

Since $\|\ell\| = 1$, this implies $\text{dist}(w, C) > R/4$. Moreover,

$$\|w - a\| < \frac{R}{4} + \frac{R}{2} + \frac{R}{8} = \frac{7R}{8} < R,$$

so the new closed ball lies in the original one. □

4 Dense Fréchet differentiability at the essential level

Theorem 1. *Suppose $m > -\infty$ and F'_m is of second category. Then the points $x \in V = \text{int } F'_m$ at which f is Fréchet differentiable with derivative zero are dense in V , and hence dense in F'_m .*

Proof. Fix any closed ball $B_0 = \overline{B}(a_0, R_0) \subset V$. Set $q = 1/8$, $R_k = R_0 q^k$, and, for $k \geq 1$,

$$\alpha_k = m - R_0 q^{2k}, \quad \delta_k = m - \alpha_k = R_0 q^{2k}.$$

Then $\alpha_k \uparrow m$. By source Theorem 3.1, F_{α_k} is nowhere dense. Hence

$$C_k := \overline{F_{\alpha_k}}$$

is closed, convex, and has empty interior.

Apply Lemma 1 inside B_{k-1} to obtain recursively a closed ball $B_k \subset B_{k-1}$, of radius R_k , such that

$$\text{dist}(B_k, C_k) > \frac{R_{k-1}}{4} = 2R_k.$$

The nested closed balls have radii tending to zero. Completeness gives a unique point $x \in \bigcap_{k \geq 0} B_k$. Since $x \notin C_k$, $f(x) \geq \alpha_k$ for every k , and therefore $f(x) \geq m$. On the other hand, $x \in B_0 \subset V \subset F'_m$, so $f(x) \leq m$. Thus $f(x) = m$.

Let $y \rightarrow x$, $y \neq x$. Eventually $y \in V$, hence $f(y) \leq m$. Choose the unique $k \geq 0$ such that

$$R_{k+1} \leq \|y - x\| < R_k.$$

As $y \rightarrow x$, this $k \rightarrow \infty$, so we may assume $k \geq 1$. Because $x \in B_k$ and $\text{dist}(B_k, C_k) > 2R_k$, the inequality $\|y - x\| < R_k$ gives $y \notin C_k$. Consequently $f(y) \geq \alpha_k$, and

$$0 \leq \frac{f(x) - f(y)}{\|x - y\|} \leq \frac{m - \alpha_k}{R_{k+1}} = \frac{R_0 q^{2k}}{R_0 q^{k+1}} = q^{k-1} \rightarrow 0.$$

Since $f(y) \leq f(x)$ near x , this is precisely $f(y) - f(x) = o(\|y - x\|)$. Thus f is Fréchet differentiable at x with derivative zero.

The initial ball $B_0 \subset V$ was arbitrary, so these points are dense in V . Finally, any convex set with nonempty interior is contained in the closure of its interior; hence V is dense in F'_m . □

5 Full answer

Theorem 2. *Every real-valued, densely continuous, quasiconvex function on a separable Banach space is continuous and classically Gâteaux differentiable at the points of a dense subset.*

Proof. If $m = -\infty$, source Corollary 4.3 gives Hadamard differentiability outside an Aronszajn-null set. Suppose $m > -\infty$. If F'_m is of first category, source Theorem 5.2 gives Hadamard differentiability at the points of a dense subset of X .

It remains to consider the second-category case. Theorem 1 gives a set D_0 of Fréchet points dense in F'_m . Source Theorem 4.2 gives Hadamard differentiability on $X \setminus F'_m$ except at the points of an Aronszajn-null set. Source Theorem 3.2(iii) says that $X \setminus F'_m$ is semi-open; as observed immediately after source Theorem 4.2, the resulting Hadamard points form a set D_1 dense in $X \setminus F'_m$. Thus $D_0 \cup D_1$ is dense in X .

Fréchet and Hadamard differentiability each imply continuity and classical Gâteaux differentiability, proving the claim. $\square$

6 Verification, novelty, and limitations

The exact-arithmetic checker performs 30,003 audits of the hole constants, distance margin, and rate identity $R_0 q^{2k} / R_{k+1} = q^{k-1}$. These checks corroborate the bookkeeping but do not replace the proof.

Four cheap run indexes and bounded exact-title, exact-question, author-plus-question, and core-keyword searches found no duplicate or later resolution through 11 August 2026. Because the decisive geometric lemma is elementary, the possibility of unindexed folklore remains; novelty confidence is moderate. Human review should focus on the strong-separation form used in Lemma 1, the distance margin in the nested construction, and the assembly of the relative-density statements from [1].

The theorem answers the stated dense classical Gâteaux question. It does not assert a residual, Aronszajn-conull, or Haar-conull set of classical Gâteaux points in the hard case, nor dense Hadamard or Fréchet differentiability on all of X .

References

- [1] P. J. Rabier, *Differentiability of quasiconvex functions on separable Banach spaces*, arXiv:1301.2852 (2013).

The aftermath of $\hat{F}_m^\dagger$ not always having empty interior is that, when $\dim X = \infty$, it remains an open question whether *every* densely continuous quasiconvex function f on a separable Banach space X is continuous and Gâteaux differentiable at the points of a dense subset. Nevertheless, in Section 9, this question is answered in the affirmative after the definition of Gâteaux differentiability is slightly altered.

A basic remark is that the concept of Gâteaux derivative at a point x continues to make sense if it is only required that the directional derivatives at x exist and are represented by a continuous linear form for some residual set of directions in the unit sphere. This suffices to define an “essential” Gâteaux derivative at x in a unique way, independent of the residual set of directions (which is not true if “residual” is replaced by “dense”). With this extended definition, the problem is resolved in Theorem 9.2. In particular, if X is reflexive and separable, every densely continuous quasiconvex function is continuous and essentially Gâteaux differentiable on X except at the points of a Haar null set.

Figure 1: Source PDF page 5: the unresolved classical Gâteaux question and the source’s essential-direction relaxation.